PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

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SCAN KODE QR ATAU KLIK DI MANA SAJA UNTUK BUKA PORTOFOLIO WEB

https://portobayu.vercel.app/

Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom



Telkom University

S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, mechanism design, and fabrication.

Electronics
Schematic design, circuit wiring, and hardware assembly.

Embedded Control
Microcontrollers, sensors, and actuator integration.

Prototyping
Physical build, functional testing, and validation workflows.

01 / PROJECTS

Engineering Work, in Practice · Selected Prototyping Showcase

Koleksi 5 proyek rekayasa fisik: Lengan Robot 5-DOF, Pendingin Surya, Robot Basket KRAI, Line Follower & Sumo Robot

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01 / FINAL PROJECT



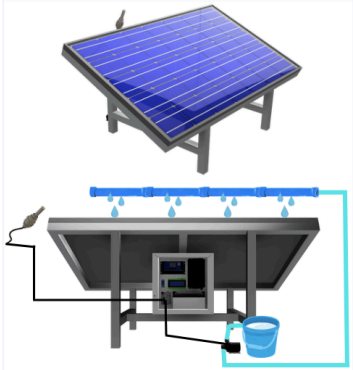
Robot Manipulator 5-DOF ↗

Prototipe lengan robot artikulasi 5-DOF bertenaga STM32F411 dengan gearbox cycloidal hasil 3D-print presisi. Mengintegrasikan encoder magnetik AS5600 closed-loop feedback untuk mereduksi backlash pada sendi artikulasi gerak, mencapai akurasi kinematika 79,8% pada beban uji 250–1.000 g.

- STM32F411 MCU
- NEMA 17 Stepper
- TB6600 Driver
- AS5600 Sensor
- TCA9548A Mux
- Cycloidal Gearb…

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02 / CAPSTONE DESIGN



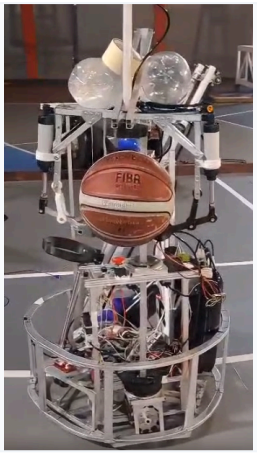
Pendingin Panel Surya Otomatis ↗

Sistem penyemprotan air otomatis untuk menurunkan temperatur operasional panel surya berbasis Arduino Uno. Memanfaatkan sensor suhu presisi dan relai pompa celup untuk memicu siklus pendinginan saat suhu >30°C dan berhenti di bawah <25°C demi memulihkan efisiensi konversi daya modul secara optimal.

- Arduino Uno R3
- RTD Temp Sensor
- 5V Relay Module
- 12V Water Pump
- 16x2 I2C LCD
- 12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



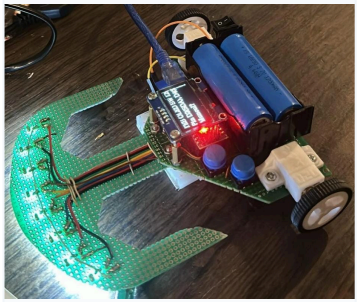
Kompetisi Robot Basket ↗

Robot penembak bola basket dinamis untuk Kontes Robot ABU Indonesia. Berperan aktif di tim mekanik merancang subsistem pelontar pneumatik 6-bar, mekanisme intake dribbler, serta chassis manuver roda omniwheel, berhasil membawa tim lolos seleksi ke Tahap Nasional ITB Jatinangor.

- 6-Bar Pneumatics
- 24V Solenoid
- RS775 DC Motor
- Omniwheel Drive
- Limit Switch & IR
- Air Regulator

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04 / PROYEK MANDIRI



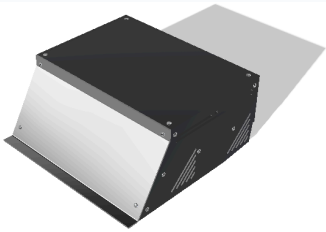
Robot Line Follower ↗

Robot pelacak garis otomatis berkecepatan tinggi dengan kemudi diferensial presisi. Dilengkapi bar sensor 8 pasang fotodioda LED, pemrosesan cepat mikrokontroler Arduino Nano, serta pengemudi motor TB6612FNG responsif dengan kalibrasi monitoring layar OLED 0.96 inci.

- Photodiode Array
- Arduino Nano
- TB6612 Driver
- N20 Gear Motor
- 0.96" I2C OLED
- 2S Li-ion Batte…

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05 / PROYEK MANDIRI



Robot Sumo ↗

Robot sumo kelas kompetisi 1 kg berdimensi 20×20 cm dengan sasis akrilik kokoh dan bilah sekop pelat aluminium. Mengandalkan 4 motor planetari PG24 torsi tinggi yang menghasilkan daya dorong 4 kg, dikendalikan nirkabel dari smartphone via Bluetooth HC-05 responsif.

- Arduino Uno
- 4x PG24 Motors
- L298N Driver
- HC-05 Bluetooth
- 3S LiPo Battery
- Acrylic Chassis

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